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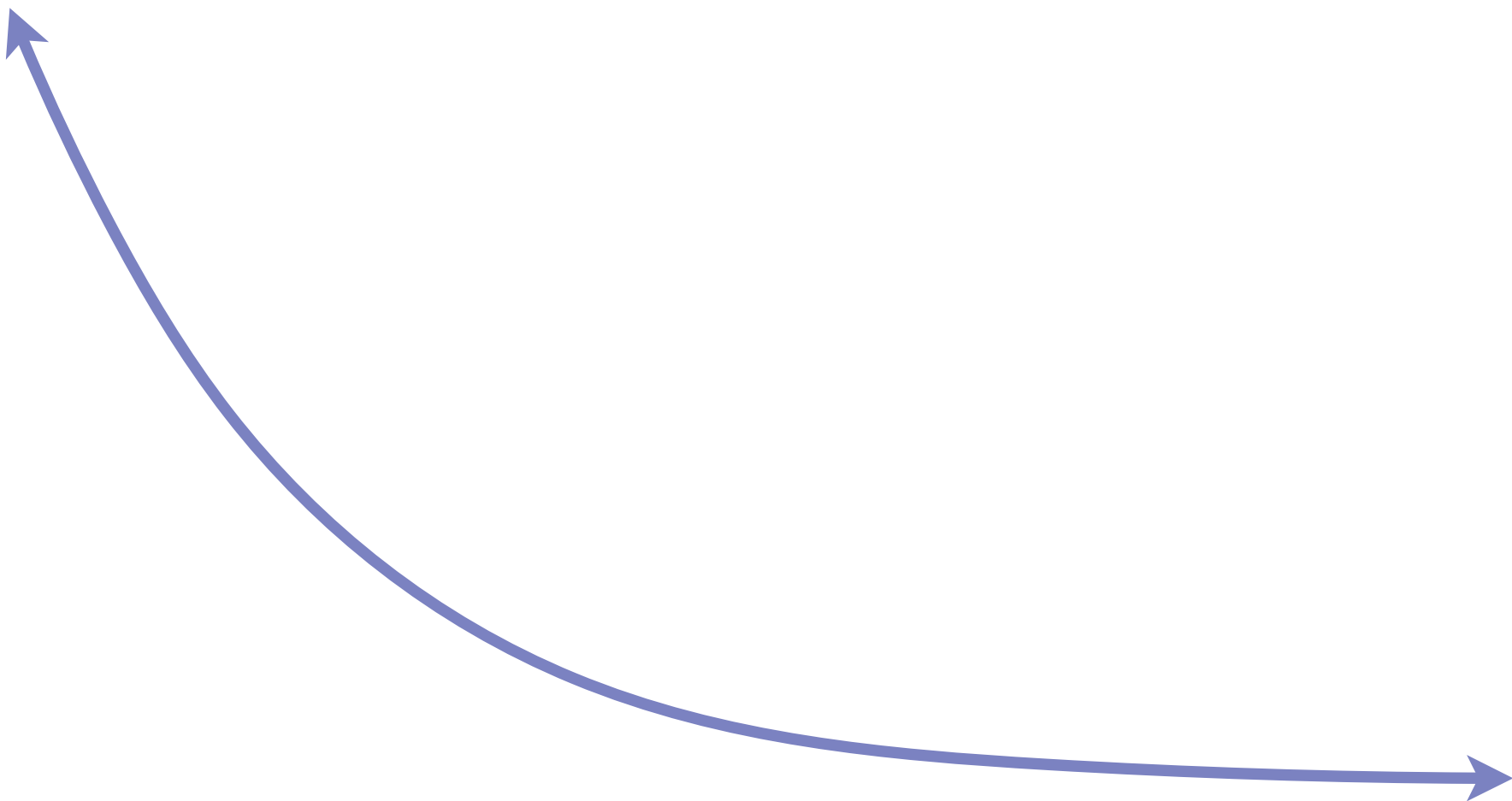
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BIAS AND VARIANCE TRADE OFF







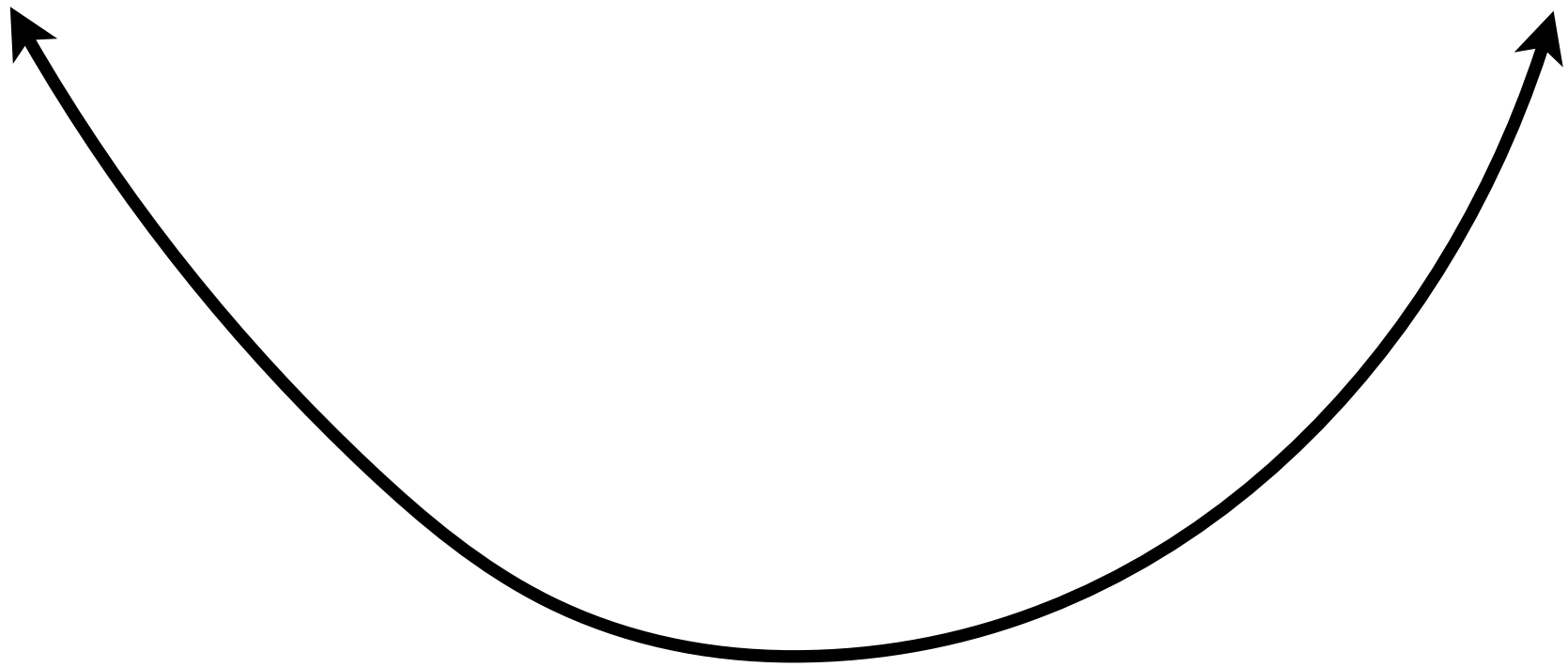


Model Complexity



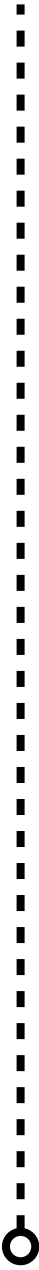
Bias

variance



Not a Error

Optimum Model Complexity



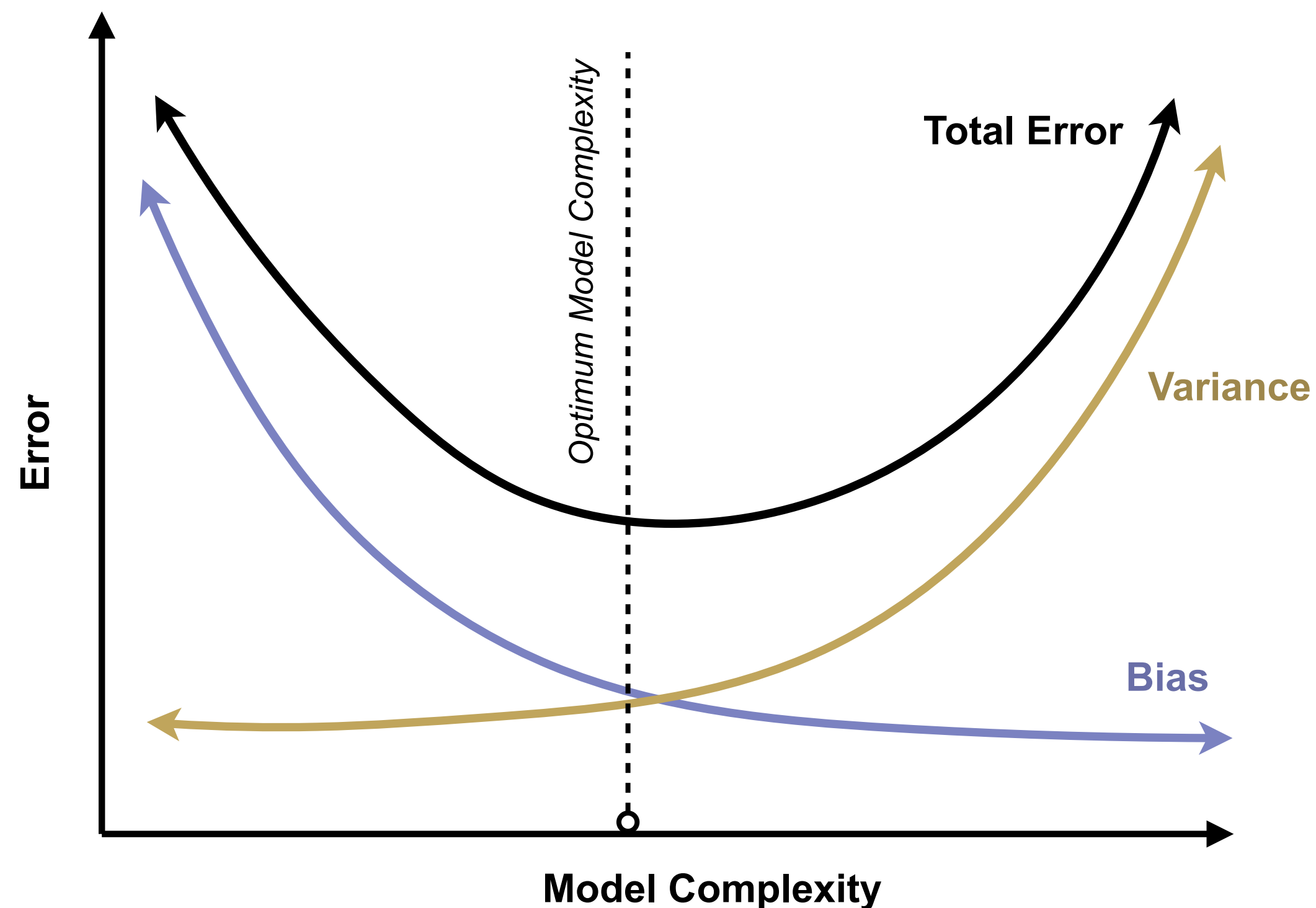
High Bias

- Strong assumptions about data form and structure.
- Simplified models may not capture complex patterns.
- Risk of underfitting: model might be too simplistic for the data.

Low Variance

- Fixed number of parameters regardless of dataset size.
- Consistent predictions with different training datasets.
- Less sensitivity to fluctuations in training data.

BIAS AND VARIANCE IN PARAMETRIC METHODS



High Bias

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K-MEANS CLUSTERING

FROM NEIGHBOURHOODS TO GROUPING!

Similar to KNN, K-means also relies on the **concept of proximity** in data space. While KNN focuses on classification based on the labels of its nearest neighbours, K-Means focuses on clustering data into groups based on their proximity to each other, **without using labels**.